

Differential Effects of Two Isoenergetic Meals Rich in Saturated or Monounsaturated Fat on Endothelial Function in Subjects With Type 2 Diabetes

OBJECTIVE—To examine the acute effects of consumption of monounsaturated (MUFAs) and saturated fatty acids (SAFAs) on endothelial function in subjects with type 2 diabetes. **RESEARCH DESIGN AND METHODS**—A total of 33 participants were examined after consumption of two different isocaloric meals: one rich in MUFA and one rich in SAFA, in the form of extra-virgin olive oil and butter, respectively. Endothelial function was assessed by determination of flow-mediated dilatation (FMD). **RESULTS**—FMD did not change significantly after the MUFA-rich meal but declined after the SAFA-rich meal. The FMD during the experiment, expressed as incremental area under the curve, increased after the MUFA-rich meal by $5.2 \pm 2.5\%$ and decreased after the SAFA-rich meal by $16.7 \pm 6.0\%$ ($\Delta = -11.5 \pm 6.4\%$; $P = 0.008$). **CONCLUSIONS**—Consumption of an SAFA-rich meal is harmful for the endothelium, while a MUFA-rich meal does not impair endothelial function in subjects with type 2 diabetes.